

Joint Trajectory Planning and Task Offloading in UAV-Assisted Inspection Networks: A Transformer-Based Approach

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Abstract—Uncrewed Aerial Vehicle (UAV) has emerged as a promising solution for automating railway inspections due to its high mobility, flexible deployment, and reduced labor cost. In this paper, we investigate UAV-assisted railway inspections, which include object recognition, humidity monitoring, and critical infrastructure modeling, each with distinct data volumes and computational requirements. Particularly, we introduce a UAV-assisted railway inspection framework. Different types of sensors are divided into several clusters. The UAV departs from the hive, flies over each cluster to collect their computational requirements, and performs task offloading before returning to the hive. This process is formulated as a joint optimization problem of trajectory planning and task offloading to minimize the weighted sum of latency and energy consumption. Considering the constrained computing and storage capabilities of UAVs, it is crucial but challenging to develop a lightweight yet high-performing solution for the multi-objective optimization problems. As such, a novel Artificial General Intelligence (AGI)-oriented Transformer (AoT) algorithm is proposed to solve the optimization problem. It uses an encoder-only architecture to process either sensor location or task features, and then directs the encoded outputs to different output heads to make decisions on UAV trajectory and task offloading. Simulation results demonstrate that the proposed AoT algorithm outperforms benchmark algorithms in terms of trajectory length and average offloading cost.

Index Terms—AGI-oriented, UAV-assisted inspection networks, shared-encoder Transformer, trajectory planning, task offloading.

I. INTRODUCTION

REPLACING traditional manual inspections with uncrewed aerial vehicle (UAV)-assisted automated one in railway industry can significantly enhance train operational safety [1]–[3]. Typical railway inspections include three aspects. First is the real-time recognition of foreign objects (debris) and unauthorized worker intrusions. Additionally, rainfall and soil moisture of mountainous areas near railway lines are needed to predict potential natural disasters. Data on critical railway infrastructure, e.g., bridges, should also be collected for shift or deformation detection. Manual inspections suffer from low efficiency and cannot promptly recognize

construction workers or infrastructure breakdowns. Moreover, collecting environmental data in mountainous areas during harsh weather can pose significant safety risks to maintenance laborers [4]–[6]. Because of its outstanding flexibility, cost-effectiveness, and ability to operate in harsh weather, UAV plays a critical role in automated railway inspections. Cameras, rain gauge, humidity sensors, and light detection and ranging (LiDAR) systems are deployed at designated locations to collect localized data [7]. UAVs periodically interact with these devices and assist in the upload and analysis of the collected data, forming UAV-assisted railway inspection networks that provide real-time safety monitoring for railway operations.

In such UAV-assisted railway inspection networks, efficient trajectory planning and task offloading are crucial for overall performance [8]–[10]. The optimized trajectory can improve the UAV working efficiency. With a fixed cruising speed, a shorter path enables the UAV to reduce total mission latency for traversing all sensor nodes. The latency for UAV recharging at the hive is also reduced accordingly. As such, the UAV can achieve a higher inspection frequency within the same period. Effective task offloading can improve the device's health status [11]. On one hand, wireless sensors deployed in mountainous areas have limited energy reservation [12]. Proper task offloading can minimize energy consumption while meeting latency requirements, thereby extending operation lifespan. On the other hand, tasks like digital twin modeling require substantial computation resources, and reasonable task offloading can prevent the rapid hardware aging by prolonged high-intensity processing [13]–[15].

With its advantages in long-range dependency modeling and efficient parallel computation, the Transformer has emerged as a promising technology for addressing the above issues [16]–[18]. By leveraging the self-attention mechanism, the Transformer can effectively capture long-range dependencies between sensor nodes [19], [20]. Thus, it is less prone to getting stuck in local optima during trajectory planning compared with traditional algorithms. Additionally, conventional learning-based task offloading algorithms typically process task features sequentially, resulting in rapidly increasing computational complexity as the problem size grows [21]. In contrast, the Transformer is inherently designed for parallel processing and is able to analyze multiple task features simultaneously. As a result, the inference time for the Transformer grows at a much slower rate, providing a significant advantage in large-scale systems.

However, achieving efficient joint trajectory planning and task offloading with the Transformer still faces several chal-

This work was supported in part by the National Key R&D Program of China under Grant 2022YFB2901900, and in part by the National Natural Science Foundation of China under Grant 62322102 and Grant 62372035. (Corresponding author: Wei Quan.)

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lenges. First, the Transformer should possess multi-task generalization capabilities [22], known as artificial general intelligence (AGI). Deploying two separate Transformer algorithms on a UAV for trajectory planning and task offloading is inefficient and consumes excessive storage resources. Therefore, a single Transformer algorithm is needed to make joint optimization. This requires the Transformer to process variable-length inputs, i.e., inputs with varying numbers of elements along with diverse feature characteristics. Second, the Transformer should also deliver satisfying performance on specific tasks while maintaining generalization capability [23]. There is always a conflict between generalization and single-task performance. To achieve balance across different tasks, the Transformer may experience a decline in single-task performance. This necessitates architectural and training innovations of the Transformer to reduce interference between various tasks. Third, the AGI-oriented Transformer algorithm needs to be lightweight for deployment on resource-constrained devices [24]. Given the limited computation resources and energy of UAVs, the Transformer should minimize the number of its model parameters to reduce both inference latency and energy consumption.

In this paper, we propose an **AGI-oriented Transformer** (AoT) algorithm for UAV-assisted automated railway inspection networks, which adopts a shared-encoder architecture to perform efficient joint trajectory planning and task offloading, aiming at reducing overall latency and energy consumption of the inspection networks. The main contributions are as follows:

- Firstly, we design a UAV-assisted railway inspection framework. The framework defines the inspection tasks, sensor types, data features, and the interaction workflow among the hive, UAV, and sensor clusters. Within the framework, an inspection delay model is built to measure the total latency required for the UAV to traverse all sensors and for the sensors to complete computation tasks. Meanwhile, an energy consumption model is established to quantify the energy consumption of UAV flight, communication, and computation, as well as the energy consumed by sensors for communication and task processing.
- Secondly, a joint optimization problem for trajectory planning and task offloading is formulated to minimize the weighted sum of overall inspection latency and total energy consumption under different resource constraints. The optimization problem is then transformed into a traveling salesman problem (TSP)-hybrid implicit Markov decision process (MDP) and an offloading MDP.
- Thirdly, we propose an AGI-oriented Transformer (AoT) algorithm to solve the joint optimization problem. The AoT algorithm adopts an encoder-only architecture to reduce the model parameters. The encoded representations from the encoder blocks are directed to different output heads as needed, including a Trajectory Planning Head and a Task Offloading Head. Instead of using a traditional decoder structure, the output heads are implemented as a multi-layer perceptron (MLP) for REINFORCE-based optimization. Simulation results validate the effectiveness

of the proposed AoT algorithm in trajectory planning and task offloading.

The remainder of this paper is organized as follows. Related works are reviewed in Section II. The framework and system model of the UAV-assisted railway inspection networks are presented in Section III. The joint optimization problem is formulated and transformed in Section IV. An AoT algorithm is developed to solve the optimization problem in Section V, followed by simulation results in Section VI. Section VII concludes this paper.

II. RELATED WORK

In recent years, extensive research has been conducted on UAV-assisted systems. A series of works focus on UAV trajectory planning to enhance UAV performance. Most works attempted to leverage new technologies to further improve the trajectory planning quality. To this end, Zhang *et al.* [25] designed a perception-aware-based UAV trajectory planner, which adopted generative adversarial self-imitation learning and class-level instance-balancing expert buffer to achieve comparable flight costs and success rates with limited planning time. In [26], a 4D trajectory planning algorithm based on the Fast Marching Square (FM2) method was designed to plan a safe and shortest path for UAVs while avoiding obstacles. In addition, for UAV-assisted inspection networks, the low coverage rate caused by unsatisfactory trajectory quality is still the main challenge. Liu *et al.* [27] aimed at maximizing UAV target coverage rate of the industrial infrastructure and reducing latency via establishing a mathematical model under energy consumption and mapping efficiency constraints. Our work differentiates from the above works in methods to improve trajectory planning quality. By combining the Transformer with REINFORCE [28], we provide an ideal training baseline for the Transformer without the need to explicitly define an MDP, resulting in faster algorithm convergence.

Another line of work focuses on efficient task offloading for UAV-assisted systems. Existing works predominantly rely on heuristic algorithms and reinforcement learning (DRL)-based methods. Tian *et al.* [29] proposed a user-centric task offloading framework for UAV-enabled MEC networks and developed a genetic algorithm (GA)-based method to maximize user satisfaction. In [30], a collaborative mobile edge computing system with multiple UAVs and multiple edge clouds was investigated, in which a multi-agent DRL framework is designed to minimize the sum of execution latency and energy consumption. However, Transformer-based task offloading solutions in UAV-assisted networks remain largely unexplored. A relevant example can be found in the connected vehicle domain. In [31], Yan *et al.* investigated an efficient offloading approach for delay-sensitive and computationally demanding tasks to ensure the safe operation of connected vehicles. Different from the above works, our paper aims to provide a lightweight model for efficient deployment on resource-constrained UAVs. By adopting an encoder-only architecture, we significantly reduce the number of parameters, which in turn lowers both the computation latency and the inference complexity during UAV deployment.

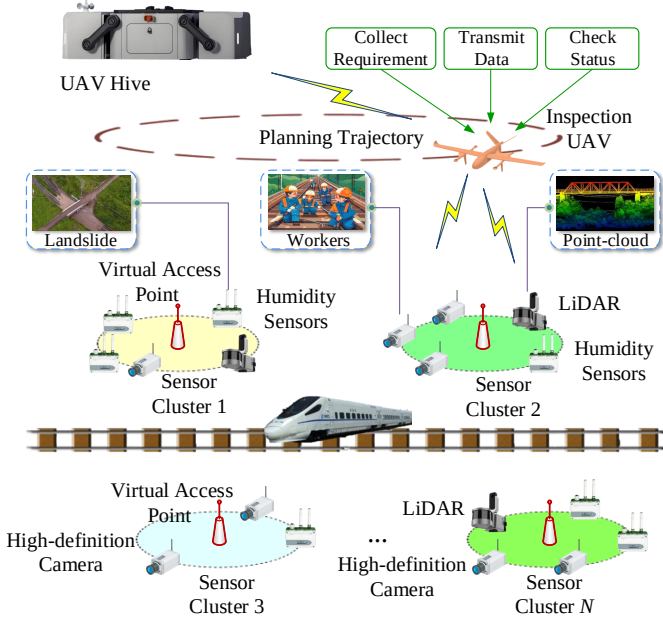


Fig. 1. The UAV-assisted inspection framework.

Many researchers devoted to using the Transformer to enhance the UAV performance since its significant advantages in long-range dependency modeling and parallel computation. Zhu *et al.* [32] proposed a combined algorithm using weighted A* and Transformer to optimize UAV trajectory planning, aiming to minimize the total age of information collected by the UAV. In [33], a Transformer-aided multi-agent hybrid action framework was studied, in which the Transformer was used to extract motion features and then provide predictive action decision-making. As such, the deep neural networks are well partitioned for collaborative inference in UAV networks. However, few Transformer-based studies attempt to jointly address both trajectory planning and task offloading. In contrast, traditional heuristic algorithms have been widely used to solve both tasks simultaneously due to their simplicity. For instance, Wang *et al.* [34] proposed a bi-criterion ant colony optimization algorithm that simultaneously handles trajectory planning and task offloading to minimize the total cost, which achieved promising results. However, unlike the works using traditional algorithms, existing Transformer-based studies can only address either trajectory planning or task offloading, with few approaches attempting to solve both problems. Different from existing works, our AoT algorithm uses a shared encoder with identical parameters to process both sensor locations and task features, while specialized output heads refine the encoded information and make optimal decisions, enabling an AGI-oriented Transformer with strong multi-task capabilities.

III. SYSTEM MODEL

In this section, we introduce the components and workflow of the UAV-assisted inspection framework. Then, the overall inspection latency and energy consumption are measured with the inspection delay model and the energy consumption model.

A. Collaborative Inspection Framework

As shown in Fig. 1, the UAV-assisted inspection framework consists of three main components: the hive, the UAV, and various types of sensors. The hive serves two primary roles: providing recharging service for the UAV and functioning as an edge cloud server [35] to process computation tasks offloaded by the UAV. The UAV serves as a communication relay and a mobile edge computing server. If a task is offloaded to the hive, the UAV will forward the sensor data to the hive. Both trajectory planning and task offloading decisions are made by the UAV. Additionally, if the task is offloaded to the UAV, it should also perform the required computation.

The three types of tasks conducted by UAV-assisted inspection networks are denoted by the set $\mathcal{K} = \{k_r, k_p, k_m\}$, where k_r represents recognizing intrusions of unauthorized worker and foreign objects, k_p denotes predicting potential natural disasters in mountainous areas along railway lines, and k_m corresponds to updating digital models of critical infrastructure like bridges and tunnels. Various types of sensors to conduct these tasks are as follows:

1) High-definition cameras: Cameras are used to detect unauthorized workers on the tracks and foreign objects intruding onto the railway (e.g., kites tangled in overhead contact lines). They typically generate high-resolution images and video frames, resulting in large data volumes. However, this data only requires simple recognition algorithms that can quickly figure out results with low computation complexity.

2) Rain gauge and humidity sensors: They are typically wireless sensors deployed in mountainous areas with limited energy. They are activated during extreme weather to collect data like rainfall and soil moisture, which will then be used to predict the likelihood of natural disasters like landslides. While the data they generate is usually measured in kilobytes, it requires complex analytical algorithms to extract meaningful patterns such as temporal sequences indicating soil saturation thresholds for natural disaster prediction.

3) LiDAR systems: This system uses LiDAR point cloud imaging to update digital models of critical infrastructure and detect deformations. Point cloud data requires substantial storage resource and must undergo a series of complex processing, including feature extraction and point cloud registration, leading to high computation complexity.

The workflow of the inspection framework is as follows. We consider a scenario where a hive h and a UAV u cover an inspection area. Different sets of sensors, including cameras $\mathcal{C} = \{1, 2, \dots, C\}$, rain gauges $\mathcal{R} = \{1, 2, \dots, R\}$, and LiDARs $\mathcal{I} = \{1, 2, \dots, I\}$, are grouped into clusters $\mathcal{N} = \{1, 2, \dots, N\}$ based on their spatial distribution. Each cluster contains a set of sensors $S_n \subseteq \mathcal{C} \cup \mathcal{R} \cup \mathcal{I}$ composed of the aforementioned sensor types. Moreover, we define a spatial coherence assumption that all sensors within a cluster are geographically close to a reference location, named the UAV virtual access point. These virtual access points $\mathcal{X} = \{x_1, x_2, \dots, x_N\}$ are selected as the center of each cluster $n \in \mathcal{N}$. Each sensor $s \in S_n$ should be located within the maximum allowable distance $D_{\max}$ from its corresponding access point x_n , expressed as:

$$\|L(s) - L(x_n)\| \leq D_{\max}, \quad s \in S_n. \quad (1)$$

TABLE I
MAIN NOTATIONS AND DEFINITIONS

Notation	Definition
$\mathcal{T}_n$	Tasks generated by sensors within the cluster n
x_n	Virtual access point of cluster n
D_{total}	Total latency including UAV flight and task offloading
E_{total}	Total energy consumption of flight and offloading
t_n^i	A task i generated in cluster n , $i \in \mathcal{T}_n$
$L(\cdot)$	Mapping function for virtual access point to its coordinate
f_l, f_u, f_h	CPU frequency of sensors, the UAV, and the hive
λ_n^i	Task offloading decision vector of t_n^i
$ h_{pu} ^2$	Link channel gain, simplified as the same for all clusters
$P_l, \bar{P}_u$	Transmission power of sensors and the UAV
$\kappa_l, \kappa_u, \kappa_h$	Effective switched capacitance of sensors, UAV, and hive

Here, $L(\cdot)$ denotes the 2D coordinates of a device and all clusters share the same D_{max} . This spatial coherence assumption offers two key modeling advantages: (1) it simplifies UAV trajectory planning by enabling cluster-level hovering instead of point-to-point navigation, and (2) it allows for a uniform representation of sensor-to-UAV transmission rate and power within each cluster, since all sensors are at roughly the same distance from the UAV during data collection.

In this framework, we consider a large-sized rotary-wing UAV platform that is widely used in inspection scenarios. Such a UAV has a flight endurance of 50 minutes to 2 hours, a cruising speed of approximately 15 m/s, and a limited payload capacity. In addition, we assume that both the UAV-sensor and UAV-hive links operate over 5G, which support parallel uplink transmissions and compatibility with existing terrestrial networks. During a single inspection, the UAV departs from the hive and plans a trajectory to visit each virtual access point. When the UAV reaches the cluster n , it will stay at the access point x_n for several minutes to collect sensor computational requirements, forward task data, and, if necessary, perform onboard computation. For each task generated within cluster n , denoted by the set $\mathcal{T}_n = \{1, 2, \dots, T_n\}$, $n \in \mathcal{N}$, the UAV determines the appropriate offloading location with a decision vector $\lambda_n^i = (\lambda_{n,i}^l, \lambda_{n,i}^u, \lambda_{n,i}^h)$, $i \in \mathcal{T}_n$. Here, each element $\lambda_{n,i}^l$, $\lambda_{n,i}^u$, and $\lambda_{n,i}^h$ is a binary indicator specifying whether the task is processed at the sensor, the UAV, or the hive, respectively. For instance, if the selected offloading location is the hive, then $\lambda_{n,i}^h = 1$, while other elements are set to 0. In this case, the sensor transmits the data to the UAV, which then forwards it to the hive. After visiting all access points, the UAV returns to the hive for recharging, marking the end of the inspection cycle.

B. Inspection Delay Model

The total latency D_{total} of the UAV-assisted inspection networks consists of the UAV flight latency D_{fly} and the task offloading latency D_{off} , where D_{off} is calculated differently depending on the offloading location. In this subsection, we first present the derivation of offloading latency $D_{\text{off}}^{n,i}$ and

energy consumption $E_{\text{off}}^{n,i}$ for a single task t_n^i , $n \in \mathcal{N}$, $i \in \mathcal{T}_n$, and the total latency and energy consumption for all tasks will be calculated at the end.

1) UAV flight latency: Let $\pi_p = \{\pi_p(0), \pi_p(1), \dots, \pi_p(n), \dots, \pi_p(N+1)\}$ denotes the UAV's visiting strategy, where each $\pi_p(n) \in \mathcal{X} \cup \{\text{hive}\}$. The total flight latency is then calculated by Eq. (2):

$$D_{\text{fly}} = \sum_{n=0}^N \frac{\|L(\pi_p(n+1)) - L(\pi_p(n))\|}{v}, \quad (2)$$

where, v represents the UAV's flying speed and is treated as a constant, and the output of $L(\cdot)$ is a 2D coordinates:

$$L(\cdot) = (X(\cdot), Y(\cdot)). \quad (3)$$

Here, $X(\cdot)$ and $Y(\cdot)$ are mapping functions that yield the horizontal and vertical coordinates of sensors or the hive, respectively. The function $\|\cdot\|$ computes the L2-norm between two nodes, representing the distance between different virtual access points. Note that the UAV departs from the hive and eventually returns to it for recharging. Therefore, the distance from the hive to the first visited point, as well as the distance from the last visited point back to the hive, should also be included in the total latency calculation.

2) Features of tasks: Before measuring the offloading delay, we need to quantify the task features relevant to latency computation. As mentioned before, this paper considers three types of tasks. Each kind of task includes three features: task type, data size and computation requirement. Let k_n^i denotes the type of a single task t_n^i , and $\mathcal{S}_n^i$ denotes the data size of each task t_n^i that under the coverage of x_n . Data from humidity sensors and LiDAR are collected periodically with nearly constant sizes and minor fluctuations. Furthermore, the data of high-definition cameras are usually segmented into standardized units with lower and upper bounds in the UAV-assisted inspection scenario. Thus, the data sizes of each type of task follow a truncated Gaussian distribution with different parameters [36], which is described as Eq. (4):

$$\mathcal{S}_n^i \sim \begin{cases} \text{TN}(\mu_1, \sigma_1^2, [g_1^{\min}, g_1^{\max}]), & \text{if } k_n^i = k_r, \\ \text{TN}(\mu_2, \sigma_2^2, [g_2^{\min}, g_2^{\max}]), & \text{if } k_n^i = k_p, \\ \text{TN}(\mu_3, \sigma_3^2, [g_3^{\min}, g_3^{\max}]), & \text{if } k_n^i = k_m. \end{cases} \quad (4)$$

Here, $\mathcal{S}_n^i$ is the size of task t_n^i . Additionally, $g_1^{\min}$, $g_2^{\min}$, and $g_3^{\max}$, etc., are constants that define the upper and lower bounds of data size for three types of tasks. Symbols μ_1 , μ_2 , and σ_1 , etc., are constants to describe the mean and variance of the Gaussian distribution for different tasks. The notation $\text{TN}(\cdot)$ represents the truncated Gaussian distribution function, defined by the following probability density function (PDF):

$$p(\mathcal{S}_n^i) = \begin{cases} \frac{1}{\sigma\sqrt{2\pi}} \cdot \frac{\exp\left(-\frac{(\mathcal{S}_n^i - \mu)^2}{2\sigma^2}\right)}{\Phi\left(\frac{g^{\max} - \mu}{\sigma}\right) - \Phi\left(\frac{g^{\min} - \mu}{\sigma}\right)}, & \text{if } \mathcal{S}_n^i \in [g^{\min}, g^{\max}], \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

Here, μ , σ , $g^{\min}$, and $g^{\max}$ have the same meanings as defined in Eq. (4). Function $\Phi(\cdot)$ denotes the cumulative distribution

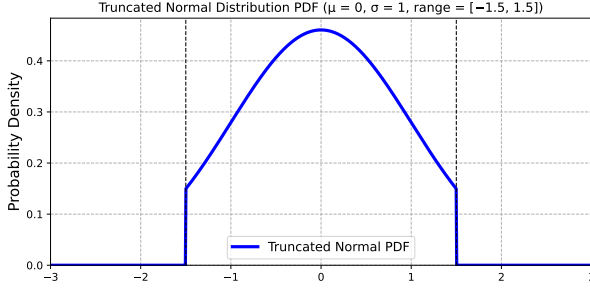


Fig. 2. Probability Density Function of the Truncated Normal Distribution.

function of the standard normal distribution. For ease of understanding, we present an illustrative PDF of $p(\mathcal{S}_n^i)$ with $\mu = 0$, $\sigma = 1$, $g^{\max} = 1.5$, and $g^{\min} = -1.5$ in Fig. 2.

In our model, while the data sizes of tasks within the same type varies, their computational requirements are assumed to be constant and are measured in central processing unit (CPU) cycles, denoted by $\mathcal{C}_r$, $\mathcal{C}_p$, and $\mathcal{C}_m$, respectively. We use $\mathcal{C}_n^i$ to denote the computation requirement of task t_n^i , which takes one of the three defined values.

3) Local offloading latency: In this scenario, the offloading latency $D_{\text{off},l}^{n,i}$ of task t_n^i is the time required to complete the computation on sensors, which is expressed as:

$$D_{\text{off},l}^{n,i} = \frac{\mathcal{C}_n^i}{f_l}, \quad (6)$$

where f_l is the CPU frequency of the sensor. We assume that three types of sensors have the same CPU frequency.

4) UAV Offloading latency: In this scenario, sensors send the data used for computation to the UAV, after which the UAV performs the computation and obtains the result. Note that the result is reported to the remote control center only when an anomaly is detected, so we only consider the latency and energy consumption of the process: sensor $\rightarrow$ UAV. Therefore, the offloading latency $D_{\text{off},u}^{n,i}$ consists of the transmission latency $D_{t,S2U}^{n,i}$ from sensors to UAV and the computation latency of the UAV $D_{c,u}^{n,i}$:

$$D_{\text{off},u}^{n,i} = D_{t,S2U}^{n,i} + D_{c,u}^{n,i}. \quad (7)$$

The computation latency of the UAV can be calculated by:

$$D_{c,u}^{n,i} = \frac{\mathcal{C}_n^i}{f_u}, \quad (8)$$

where f_u is the CPU processing frequency of UAV. The data transmission latency can be calculated by:

$$D_{t,S2U}^{n,i} = \frac{\mathcal{C}_n^i}{R^{S2U}}, \quad (9)$$

where R^{S2U} is the transmission rate between the sensor and the UAV, which is regarded the same for all sensors and is calculated by:

$$R^{S2U} = B_{pu} \times \log_2 \left(1 + \frac{P_t |h_{pu}|^2}{N_0 B_{pu}} \right). \quad (10)$$

Here, B_{pu} is the uplink bandwidth between a sensor and the UAV, and we assume that the bandwidth between all

sensors and the UAV are all the same. P_t represents the sensor transmission power, it is also regarded as the same for all sensors. N_0 is the channel noise power spectral density for the UAV. The link channel gain $|h_{pu}|^2$ between all sensors and the UAV is derived by:

$$|h_{pu}|^2 = \beta_0 \times H^{-\alpha}, \quad (11)$$

where β_0 is path loss coefficient at the reference distance (typically 1 meter), H is the height of UAV, and α is the shared path loss exponent.

5) Hive offloading latency: In this scenario, data flows in the order of sensor $\rightarrow$ UAV $\rightarrow$ hive. Therefore, the offloading latency $D_{\text{off},h}^{n,i}$ consists of the transmission latency from the sensor to UAV $D_{t,S2U}^{n,i}$, the latency from the UAV to hive $D_{t,U2H}^{n,i}$, and the computation latency at the hive $D_{c,h}^{n,i}$,

$$D_{\text{off},h}^{n,i} = D_{t,S2U}^{n,i} + D_{t,U2H}^{n,i} + D_{c,h}^{n,i}. \quad (12)$$

Here, the transmission latency from the UAV to the hive $D_{t,U2H}^{n,i}$ is calculated by:

$$D_{t,U2H}^{n,i} = \frac{\mathcal{S}_n^i}{R^{U2H}}, \quad (13)$$

where we assume that the transmission rate between the UAV and the hive R^{U2H} remains constant across different clusters. The computation latency $D_{c,h}^{n,i}$ is described as:

$$D_{c,h}^{n,i} = \frac{\mathcal{C}_n^i}{f_h}, \quad (14)$$

where f_h is the CPU frequency of the hive. As such, we derive the offloading latency of a single task t_n^i .

C. Energy Consumption Model

The energy consumption of the inspection networks E_{total} includes the UAV flight consumption E_{fly} and the task offloading consumption E_{off} , which is calculated as follows.

1) UAV flight consumption: Since the hovering time of the UAV at each cluster is fixed and not subject to optimization, we focus solely on the energy consumption E_{fly} incurred during its cruising, which is described as:

$$E_{\text{fly}} = \int_0^T P_{\text{fly}}(v(t)) dt \approx P_{\text{fly}}(v) \times \frac{\mathcal{L}}{v} = P_{\text{fly}}(v) \times D_{\text{fly}}. \quad (15)$$

Here, $\mathcal{L}$ denotes the trajectory length, calculated as $\sum_{n=0}^N ||L(\pi_p(n+1)) - L(\pi_p(n))||$. The UAV's flight power P_{fly} is a function of speed [37]. For simplification, we assume the UAV flies at a fixed speed, thus $P_{\text{fly}}(v)$ can be derived by:

$$P_{\text{fly}} = \frac{1}{2} \cdot C_d \cdot \rho \cdot A \cdot v^3, \quad (16)$$

where C_d denotes the drag coefficient, ρ is the air density, and A represents projected area of the UAV facing the airflow.

2) Local offloading consumption: In this scenario, $E_{\text{off},l}^{n,i}$ is given by Eq. (17):

$$E_{\text{off},l}^{n,i} = \kappa_1 \times \mathcal{C}_n^i \times f_l^2, \quad (17)$$

where κ_1 is the effective switched capacitance shared by all sensors, and $\mathcal{C}_n^i$ takes one of the values from $\mathcal{C}_r$, $\mathcal{C}_p$, and

$\mathcal{C}_m$. Moreover, Eq. (17) can be combined with Eq. (6) and rewritten as:

$$\begin{aligned} E_{\text{off},l}^{n,i} &= \kappa_l \mathcal{C}_n^i f_1^2 \\ &= \kappa_l \mathcal{C}_n^i \left(\frac{\mathcal{C}_n^i}{D_{\text{off},l}^{n,i}} \right)^2 \\ &= \kappa_l \frac{(\mathcal{C}_n^i)^3}{(D_{\text{off},l}^{n,i})^2}. \end{aligned} \quad (18)$$

As such, the energy consumption for local computing can be directly obtained from the result of Eq. (6).

3) UAV offloading consumption: The energy consumption for computing on the UAV $E_{\text{off},u}^{n,i}$ consists of the transmission energy consumption $E_{\text{t},\text{S2U}}^{n,i}$ of the sensor and the computation consumption $E_{\text{c},u}^{n,i}$ of the UAV,

$$\begin{aligned} E_{\text{off},u}^{n,i} &= E_{\text{t},\text{S2U}}^{n,i} + E_{\text{c},u}^{n,i} \\ &= \frac{\mathcal{S}_n^i}{R_{\text{S2U}}} \cdot P_l + \kappa_u \times \mathcal{C}_n^i \times f_u^2 \\ &= \frac{\mathcal{S}_n^i}{R_{\text{S2U}}} \cdot P_l + \kappa_u \frac{(\mathcal{C}_n^i)^3}{(D_{\text{c},u}^{n,i})^2}. \end{aligned} \quad (19)$$

Here, κ_u is the effective switched capacitance of the UAV. Based on the receiver sensitivity of UAV P_{th}^u and the channel gain defined in Eq. (11), the required transmission power P_l of the sensor can be expressed as:

$$P_l = \frac{P_{\text{th}}^u}{|h_{\text{pu}}|^2} = \frac{P_{\text{th}}^u}{\beta_0 \times H^{-\alpha}} = P_{\text{th}}^u \cdot \frac{H^\alpha}{\beta_0}. \quad (20)$$

By substituting the value of P_l into Eq. (19), the offloading consumption $E_{\text{off},u}^{n,i}$ can be obtained.

4) Hive offloading consumption: When the computation is performed at the hive, the energy consumption $E_{\text{off},h}^{n,i}$ includes the transmission energy of the sensor $E_{\text{t},\text{S2U}}^{n,i}$ and the transmission energy of UAV $E_{\text{t},\text{U2H}}^{n,i}$, as well as the computation energy at the hive $E_{\text{c},h}^{n,i}$. Given the effective switched capacitance of the hive κ_h , the $E_{\text{off},h}^{n,i}$ is calculated as:

$$\begin{aligned} E_{\text{off},h}^{n,i} &= E_{\text{t},\text{S2U}}^{n,i} + E_{\text{t},\text{U2H}}^{n,i} + E_{\text{c},h}^{n,i} \\ &\approx \frac{\mathcal{S}_n^i}{R_{\text{S2U}}} \cdot P_l + \frac{\mathcal{S}_n^i}{R_{\text{U2H}}} \cdot \bar{P}_u + \kappa_h \times \mathcal{C}_n^i \times f_h^2 \\ &= \frac{\mathcal{S}_n^i}{R_{\text{S2U}}} \cdot P_{\text{th}}^u \cdot \frac{H^\alpha}{\beta_0} + \frac{\mathcal{S}_n^i}{R_{\text{U2H}}} \cdot \bar{P}_u + \kappa_h \frac{(\mathcal{C}_n^i)^3}{(D_{\text{c},h}^{n,i})^2}. \end{aligned} \quad (21)$$

As mentioned before, R_{U2H} is treated as a constant, which requires the UAV to dynamically adjust its transmission power at different clusters. In this paper, we reasonably simplify this variability by using an average value. Thus, the average transmission power $\bar{P}_u$ is described as:

$$\bar{P}_u = \frac{P_{\text{th}}^h}{|h_{\text{uh}}|^2} = \frac{P_{\text{th}}^u}{\beta_1 \times \bar{d}^{-\alpha}} = P_{\text{th}}^u \cdot \frac{\bar{d}^\alpha}{\beta_1}. \quad (22)$$

where P_{th}^h is the receiver sensitivity of the hive, and β_1 is the path loss coefficient under near line-of-sight (LoS) conditions for the UAV. In addition, we use the distance from the center of the inspection area to the hive to represent the UAV's average communication distance, denoted as $\bar{d}$.

D. Total Latency and Energy Consumption

The total latency D_{total} is calculated by summing the UAV flight latency D_{fly} and the offloading latency $D_{\text{off}}^{n,i}$ of all tasks across all clusters. The total energy consumption E_{total} can be obtained in a similar manner, described as:

$$D_{\text{total}} = D_{\text{fly}} + \sum_{n=1}^N \sum_{i=1}^{T_n} \left(\lambda_{n,i}^l D_{\text{off},l}^{n,i} + \lambda_{n,i}^u D_{\text{off},u}^{n,i} + \lambda_{n,i}^h D_{\text{off},h}^{n,i} \right), \quad (23)$$

$$E_{\text{total}} = E_{\text{fly}} + \sum_{n=1}^N \sum_{i=1}^{T_n} \left(\lambda_{n,i}^l E_{\text{off},l}^{n,i} + \lambda_{n,i}^u E_{\text{off},u}^{n,i} + \lambda_{n,i}^h E_{\text{off},h}^{n,i} \right). \quad (24)$$

Eqs. (23) and (24) bridge the system model with the optimization problem, which depicts how the modeling results of a single task are used to calculate the overall latency and energy consumption.

IV. PROBLEM FORMULATION AND TRANSFORMATION

In this section, we formulate an optimization problem to make trajectory planning and offloading decisions and decompose it into a traveling salesman problem (TSP)-hybrid implicit Markov decision process (MDP) and an offloading MDP.

A. Problem Formulation

Let π_p denote the UAV's visiting strategy of virtual access points and the hive, $\pi_o = \{\lambda_n^i\}$ denote the offloading decision for each task t_n^i . The goal of the optimization problem is to minimize the weighted sum of total latency and energy consumption subject to a set of constraints,

$$\begin{aligned} P_0 : \quad & \min_{\pi_p, \pi_o} \quad \omega_D \cdot D_{\text{total}} + \omega_E \cdot E_{\text{total}} \\ \text{s.t.} \quad & \pi_p(n) \in \mathcal{X} \cup \{\text{hive}\}, \quad \pi_p(0) = \pi_p(N+1) = \text{hive}, \end{aligned} \quad (25a)$$

$$\lambda_{n,i}^l, \lambda_{n,i}^u, \lambda_{n,i}^h \in \{0, 1\}, \quad \forall n \in \mathcal{N}, i \in \mathcal{T}_n, \quad (25b)$$

$$\lambda_{n,i}^l + \lambda_{n,i}^u + \lambda_{n,i}^h = 1, \quad \forall n \in \mathcal{N}, i \in \mathcal{T}_n, \quad (25c)$$

$$\sum_{n=1}^N \sum_{i=1}^{T_n} \lambda_{n,i}^u E_{\text{c},u}^{n,i} \leq E_{\text{c},u}^{\text{max}}, \quad (25d)$$

$$D_{\text{off}}^{n,i} \leq D_{\text{off}}^{\text{max}}, \quad \forall n \in \mathcal{N}, i \in \mathcal{T}_n, \quad (25e)$$

Constraint (25a) ensures the UAV to complete a closed-loop trajectory, and visit each cluster exactly once. Additionally, each task can only be completely offloaded to one location which is ensured by constraints (25b) and (25c). Constraint (25d) imposes a computation energy budget on the UAV, while (25e) guarantees that the offloading latency of any task does not exceed the maximum latency tolerance. Since energy (joules) and latency (seconds) differ in units and scales, we adopt a weighted sum formulation, in which the weights ω_D and ω_E normalize their impacts on the optimization problem.

B. Problem Transformation

The flight latency and energy consumption of the UAV are relatively independent, and the visiting strategy π_p does not influence the task offloading decisions in problem P_0 . Thus, the trajectory planning sub-problem can be transformed into a TSP.

$$\begin{aligned} P_1 : \quad & \min_{\pi_p} \sum_{n=0}^N \|\pi_p(n) - \pi_p(n+1)\| \\ \text{s.t.} \quad & (25a). \end{aligned} \quad (26a)$$

To guarantee the performance of the proposed AoT algorithm, we integrate the REINFORCE algorithm with the Transformer architecture and formulate a TSP-hybrid implicit MDP, which enhances the generalization and exploration capabilities. The goal of the hybrid TSP-MDP is to minimize the tour length via self-attention-based policy learning,

$$\begin{aligned} P_1^m : \quad & \max_{\pi_p^\theta} \mathbb{E}_{\pi_p^\theta} \left[\sum_{n=0}^N \log \pi_p^\theta(a_n^p | s_n^p) \cdot r_n^p \right] \\ \text{s.t.} \quad & (25a). \end{aligned} \quad (27a)$$

Here, π_p^θ denotes the trajectory planning strategy generated by a neural network with a set of parameters θ , and r_n^p is defined as the negative value of the path length,

$$r_n^p = -\|\pi_p^\theta(n) - \pi_p^\theta(n+1)\|. \quad (28)$$

Unlike traditional MDPs, the state in the hybrid TSP-MDP is implicitly represented by the contextual embedding of the Transformer over locations and visited points. Additionally, the action is the selection of the next visiting access point $a_n^p \in \{x_1, x_2, \dots, x_N, \text{hive}\} \setminus \{a_0^p, a_1^p, \dots, a_{n-1}^p\}$.

The objective of the task offloading MDP is to minimize the weighted sum of delay and energy consumption across all tasks, which can be expressed as:

$$\begin{aligned} P_2^m : \quad & \max_{\pi_o^{\theta'}} \mathbb{E}_{\pi_o^{\theta'}} \left[\sum_{n=1}^N \sum_{i=1}^{T_n} \log \pi_o^{\theta'}(a_n^i | s_n^i) \cdot r_n^i \right] \\ \text{s.t.} \quad & (25b) - (25e). \end{aligned} \quad (29a)$$

Here, $\pi_o^{\theta'}$ denotes the offloading decision generated by a neural network with parameters θ' , and the reward r_n^i is defined as the negative weighted sum of the offloading latency and energy consumption or an offloading failure penalty of task t_n^i , described as:

$$r_n^i = \begin{cases} -(\omega_D D_{\text{off}}^{n,i} + \omega_E E_{\text{off}}^{n,i}), & \text{if } a_n^i \text{ is valid,} \\ -1.2 \times (\omega_D D_{\text{off}}^{n,i} + \omega_E E_{\text{off}}^{n,i}), & \text{otherwise.} \end{cases} \quad (30)$$

The state consists of feature vectors of task t_n^i and UAV computation energy budget,

$$s_n^i = \left(k_n^i, \mathcal{C}_n^i, \mathcal{S}_n^i, e_{\text{bgt}}^{n,i} \right), \quad (31)$$

where k_n^i denotes the type of task t_n^i and $e_{\text{bgt}}^{n,i}$ is the remaining budget for UAV computation. The action λ_n^i has been introduced before.

V. AoT: AGI-ORIENTED TRANSFORMER ALGORITHM

In this section, we propose the AoT algorithm for making joint trajectory planning and task offloading decisions. We first present the architecture design of the AoT algorithm. Then, we describe the training procedure of AoT, followed by an analysis of the algorithm computational complexity.

A. Architecture Design of the AoT Algorithm

As illustrated in Fig. 3, the architecture of the AoT algorithm consists of three main components: (1) variable-length input embedding, (2) shared encoder for trajectory planning and task offloading, and (3) dual output heads responsible for task-specific decision-making. In each component, we adopt specific designs to ensure that the algorithm remains lightweight while maintaining satisfying performance. The detailed description of each component is as follows.

1) Input Embedding: The AoT algorithm accepts two distinct types of input depending on the task. For trajectory planning, the input is a sequence of 2D coordinates representing the virtual access points of sensor clusters. These coordinates are linearly projected into high-dimensional vectors of size d using fully connected embedding layers. Note that the Transformer's hidden dimension is set to d as well, i.e., the Q , K , and V matrices also have d dimensions. As shown in the upper left of Fig. 3, to reduce gradient variance and discover shorter trajectories within the same number of training episodes, we duplicate the coordinate sequence for B_p times and form a training batch. This design yields B_p parallel tours per training episode, allowing the model to explore multiple optimal or near-optimal trajectories simultaneously.

For task offloading, the input consists of the task type, features and UAV computation budget. The task type is embedded using a learnable embedding table, while others are processed by a linear layer. To prevent gradient explosion and mitigate the dominance of any single feature during training, all features are first normalized and then concatenated to form the final input for the encoder. Additionally, we split the long sequence of tasks into multiple shorter sequences to form batches, taking advantage of AoT's capability to process all sequences within a batch in parallel, thereby improving computational efficiency.

2) Shared encoder: The core of our AoT algorithm is a shared Transformer encoder that encodes coordinates of access points and features of tasks with the same model parameters. The encoder consists of several stacked TransformerEncoder layers (our model adopts 3 layers), each composed of a multi-head self-attention (MHSA) module followed by a feedforward network (FFN) module. In the MHSA module, the attention operation computes contextualized representations by attending to all other tokens in the sequence. This captures global dependencies among locations (trajectory planning) or tasks (task offloading). In the FFN module, the output of the MHSA module passes through two linear layers and a ReLU activation to enhance nonlinear feature transformation. The three identical encoder layers process the sequence hierarchically, each layer has an identical architecture with separate parameters.

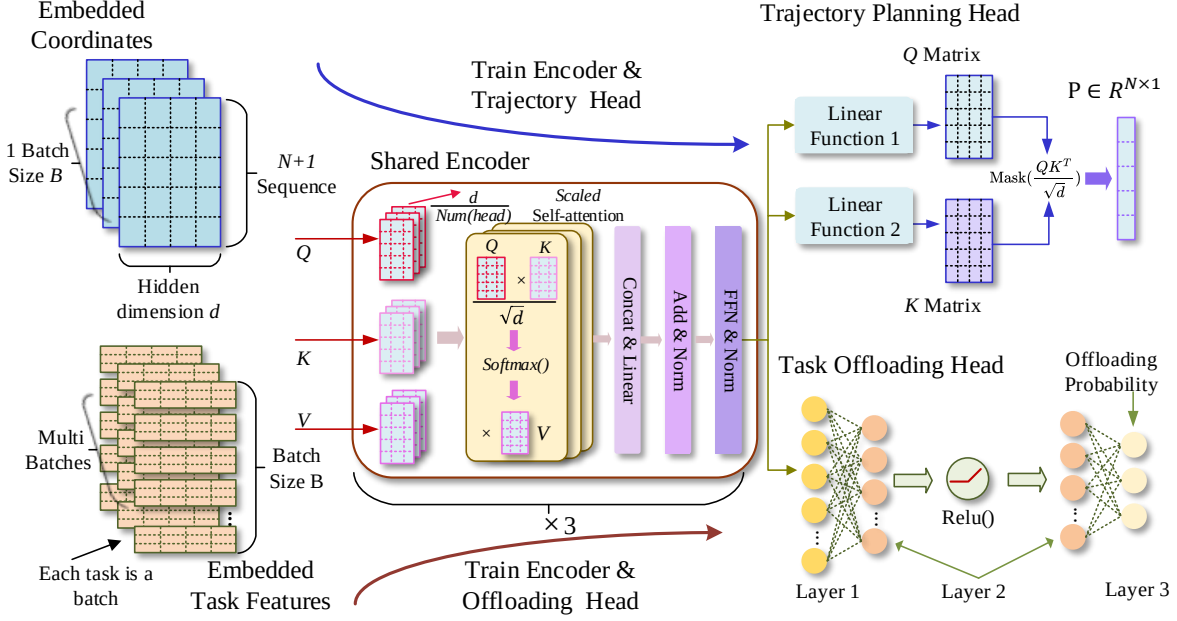


Fig. 3. The architecture design of the AoT algorithm.

The final encoder output is a batch of sequences and is then forwarded to the corresponding output head.

3) **Dual Output Heads:** To adapt to the shared encoder representations for two distinct decision-making tasks, we add two independent output heads following the shared encoder. The trajectory planning head is designed to solve the TSP in an autoregressive manner. It takes the encoder output and applies an additional set of Q and K matrices to compute the adjacency score matrix $\mathcal{A}$, which encodes pairwise transition scores between access points. To ensure valid point selection, the trajectory output head employs a masking mechanism that dynamically suppresses already visited nodes by assigning large negative values to their logits. Finally, it generates an attention score matrix using the $\text{softmax}()$ function, which serves as the basis for sampling during training.

The offloading head is implemented as a two-layer multi-layer perceptron (MLP). It takes the output of the encoder as input and passes through the MLP in the following way:

$$z_n^i = W_2(\text{ReLU}(W_1 h_n^i + b_1) + b_2, n \in \mathcal{N}, i \in \mathcal{T}_n, \quad (32)$$

where h_n^i is the final encoded representation of task t_n^i by the encoder. The weights and bias of the first and second layer are denoted by W_1, W_2, b_1, b_2 , respectively. The final output logits $z_n^i \in \mathbb{R}^3$ correspond to the three possible offloading choices: sensors, the UAV, and the hive. It define a categorical distribution from which offloading actions are sampled.

In summary, each output head is trained independently using task-specific rewards, while benefiting from the shared encoder's global representation capacity. This decoupled yet coordinated design allows the AoT algorithm to handle multiple tasks, showing the potential of AGI.

B. Training Procedure of the AoT Algorithm

In this subsection, we illustrate the training procedure of the proposed AoT algorithm in detail, shown as Algorithm 1.

1) **Initialization (Line 1):** All learnable parameters of the AoT algorithm, including the encoder parameters θ_e , the trajectory planning head θ_t , and the task offloading head θ_o are initialized in this phase. The aforementioned parameters set θ corresponds to $\theta_e + \theta_t$, and θ' corresponds to $\theta_e + \theta_o$. Additionally, the UAV's computational budget, $e_{\text{bgt}}^u = E_{u,\text{cpt}}^{\text{max}}$, and Transformer hyperparameters (e.g., number of layers, hidden dimension d) are also configured.

2) **Phase I (Lines 2-8):** This phase trains the encoder and trajectory head by combining the REINFORCE algorithm. The UAV's trajectory is a closed loop starting and ending at the hive. The initial sequence is duplicated B times with random shuffling to construct a training batch, enabling the model to explore multiple candidate trajectories per episode. The coordinate sequences are encoded using the Transformer encoder, where contextualized embeddings are computed as:

$$h_m^f = \mathcal{M}(Q \leftarrow h_m^{f-1}, K, V \leftarrow h_{1:M}^{f-1}). \quad (33)$$

Here h_m^f denotes the hidden state of the m -th token at the f -th layer of the Transformer encoder and $\mathcal{M}(\cdot)$ denotes the multi-head attention mechanism. The query vector Q is obtained by applying a linear projection to the hidden representation h_m^{f-1} of the m -th token from the previous encoder block layer, while the key and value matrices K and V are derived from the representations of all M tokens $h_{1:M}^{f-1}$. The output h_m^f is then processed by an FFN equipped with residual connection, layer normalization, and nonlinear activation. Let $\text{LyNm}(\cdot)$ denote the standard layer normalization operation and $\text{ReLU}(\cdot)$ denote the rectified linear unit activation function. The FFN operation

Algorithm 1: Training Procedure of the AoT Model

Input: Access Points $\{x_1, \dots, x_N\}$, task set $\{t_1^1, \dots, t_n^i, \dots, t_N^1\}$, UAV remaining computation budget e_{bgt}^u , and encoder settings;

Output: The optimal trajectory $\mathcal{N}_p^*$ and offloading decision for each task $\{\lambda_n^i\}$;

- 1 **Initialize:** encoder parameters θ_e , parameters of trajectory planning head θ_t and offloading head θ_o ;
- 2 **Phase I: Trajectory Planning Training;**
- 3 **for** $episode = 1$ **to** E_1 **do**
- 4 Set initial sequence $\{\text{hive}, x_1, \dots, x_N, \text{hive}\}$; Repeat the initial sequence B times to form a batch, shuffle $\{x_0, \dots, x_N\}$ for each sequence; Generate contextual representations for each sequence based on Eqs. (33) (34), and compute transition logits based on Eq. (35);
- 5 Sample tours via REINFORCE and compute tour rewards based on Eq. (36);
- 6 Update θ_e, θ_t with loss $\mathcal{L}_p$ based on Eq. (37);
- 7 **end**
- 8 Save parameters of encoder θ_e^1 and trajectory head θ_t ;
- 9 **Phase II: Task Offloading Training;**
- 10 **for** $episode = 1$ **to** E_2 **do**
- 11 **foreach** $task\ batch$ **do**
- 12 Normalize task features and UAV computation budget via Eq. (38);
- 13 Compute logits based on Eq. (32);
- 14 Sample action $a_n^i \sim \pi_o^{\theta'}(a_n^i | s_n^i)$;
- 15 **if** $a_t = \text{UAV and remaining } e_{\text{bgt}}^u > 0$ **then**
- 16 subtract the corresponding energy consumption to e_{bgt}^u and obtain reward r_n^i ;
- 17 **end**
- 18 **else**
- 19 Apply penalty to reward via Eq. (30);
- 20 **end**
- 21 Update θ_e, θ_o with $\mathcal{L}_o$ based on Eq. (39);
- 22 **end**
- 23 **end**
- 24 Save encoder parameters θ_e^2 and offloading head θ_o ;
- 25 **Phase III: Joint Parameters Optimizing;**
- 26 encoder parameters $\theta_e^3 = \varepsilon \theta_e^1 + (1 - \varepsilon) \theta_e^2$, freeze θ_e^3 ;
- 27 **for** $episode = 1$ **to** E_3 **do**
- 28 Repeat **Phase I** and **Phase II**, update θ_t and θ_o ;
- 29 **end**

is formulated as:

$$h_m^f = \text{LyNm}(h_m^f + W_2^f \cdot \text{ReLU}(W_1^f h_m^f + b_1^f) + b_2^f), \quad (34)$$

where W_1^f , W_2^f , b_1^f , and b_2^f are the learnable parameters of the FFN in the f -th Transformer encoder layer. The final encoder outputs h_n^f derived from h_m^f step by step are fed into the trajectory head, where another set of Q and K matrices compute the adjacency score matrix. A masking mechanism is applied to exclude already visited nodes, and a softmax()

function generates the action distribution,

$$\mathbf{P} = \text{Softmax}_{\text{row}}(\text{Mask}(\frac{QK^T}{\sqrt{d}})), \quad (35)$$

Here, $\text{Mask}(\cdot)$ denotes the operation of assigning a large negative value (e.g., -10^6) to already visited nodes, while $\text{Softmax}_{\text{row}}(\cdot)$ indicates applying the softmax function to a specific row of the score matrix. The next access point is sampled at each step according to the probability matrix $\mathbf{P}$, forming a full trajectory. The total reward R^b for each trajectory is calculated as:

$$R^b = \sum_{n=0}^N r_n^p. \quad (36)$$

The loss is calculated by Eq. (37), where $\hat{R}^b$ denotes the highest reward within the current batch and $\pi_p^{\theta, b}$ represents the entire trajectory generated by the policy π_p^{θ} for the b -th instance in the training batch,

$$\mathcal{L}_p(\theta) = -\frac{1}{B} \sum_{b=1}^B \left[\left(\sum_{n=0}^N \log \pi_p^{\theta, b}(n) \right) \cdot (R^b - \hat{R}^b) \right]. \quad (37)$$

In each iteration, parameters are updated via gradient descent. After E_1 episodes of training, the parameters of the encoder and the trajectory head are saved.

3) *Phase II (Lines 9-24)*: This phase focuses on training the offloading head. For each task t_n^i , computation requirement and data size, are normalized using min-max scaling:

$$x^{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}. \quad (38)$$

Here, x can denote either of them, which are scaled to the range $(0, 1)$. After estimating the probability distribution and sampling actions, the offloading reward r_n^i for each task is obtained. The model parameters are updated by minimizing the loss $\mathcal{L}_o(\theta')$,

$$\mathcal{L}_o(\theta') = -\frac{1}{\sum_{n=1}^N T_n} \sum_{n=1}^N \sum_{i=1}^{T_n} (\log \pi_o^{\theta'}(a_n^i | s_n^i) r_n^i). \quad (39)$$

After training, the learned parameters of the encoder and the offloading head are stored for final fine-tuning.

4) *Phase III (Lines 25-29)*: In this phase, we fuse encoder parameters obtained from both tasks. The final encoder parameters are linearly combined, as shown in Line 26, where $\varepsilon \in [0, 1]$ balances the influence from the trajectory and offloading training. The combined encoder is frozen, and only the output heads θ_t and θ_o are updated in this phase. This joint fine-tuning strategy enhances cross-task generalization.

C. Computational Complexity of AoT Algorithm

In this subsection, we analyze the computational complexity of the AoT algorithm during inference. The computational complexity is evaluated separately for trajectory planning and task offloading. We assume that the encoder consists of F layers with an embedding dimension of d . Each FFN has a hidden dimension of $d/2$.

TABLE II
SIMULATION AND AOT MODEL PARAMETERS

Parameter	Value	Parameter	Value
Sensor CPU f_i	0.5 GHz	Batch size	64
UAV CPU f_u	1.5 GHz	Episodes	500
Hive CPU f_h	2.0 GHz	learning rate l_r	10^{-4}
κ_l of sensors	2×10^{-25}	Encoder layers	3
κ_u of UAV	2×10^{-26}	Attention heads	8
κ_h of hive	1.25×10^{-27}	Dimension d	128
R^{S2U}	5 Mbps	MLP layers	3
R^{U2H}	30 Mbps	MLP neurons	[128, 64, 3]

For inference of trajectory planning, the batch size is set to 1. Thus, the complexity of the embedding operation is $\mathcal{O}(N \times d)$. In a single-layer Transformer encoder, generating the Query Q , Key K , and Value V matrices requires $\mathcal{O}(N \times d^2)$. The attention mechanism, including attention score computation and weighted summation, costs $\mathcal{O}(N^2 \times d)$, while the two linear transformations in the FFN layer contribute $\mathcal{O}(N \times d^2)$. With F layers, the overall computational complexity is $\mathcal{O}(F \times (2N^2d + 2Nd^2))$. The computational complexity of the trajectory planning head includes the cost of generating the Q and K matrices and their multiplication, amounting to $\mathcal{O}(N^2d + Nd^2)$. Therefore, the overall computational complexity for trajectory planning is $\mathcal{O}((2F + 1)(N^2d + Nd^2)) \approx \mathcal{O}(N^2)$, with the dominant term arising from the attention score computation and matrix multiplications.

Next, we analyze the computational complexity of task offloading inference. Assume that cluster n generates T_n tasks and the UAV makes offloading decisions for B_s tasks in parallel. Based on earlier analysis, the computational complexity of making decisions for B_s tasks with a F -layer encoder is $\mathcal{O}(F \times B_s d^2)$. The Offloading Head, consisting of two fully connected layers, adds $\mathcal{O}(B_s d^2)$. Hence, the total complexity for all tasks is $\sum_{n=1}^N \lceil \frac{T_n}{B_s} \rceil \times \mathcal{O}((F+1)B_s d^2) \approx \mathcal{O}(\sum_{n=1}^N T_n)$. In addition, only N and T_n are variables, other parameters are considered constants.

VI. SIMULATION RESULTS

In this section, extensive simulations are conducted to verify the performance of the proposed AoT algorithm for trajectory planning and task offloading.

A. Simulation Setup

We consider a scenario where a UAV departs from a hive located at the origin $(0, 0)$ and visits $[20, 30, 40, 50]$ clusters of sensors randomly distributed within a 1000×1000 meter area, and then returns to the hive for recharging. A total of 488 tasks are generated following Poisson and truncated Gaussian distributions. Specifically, target recognition tasks are sampled with an average of 125 tasks and data sizes ranging from 3 to 5 MB, risk prediction tasks average 300 with data sizes between 0.9 to 1.1 MB, and LiDAR modeling tasks average 75 with sizes from 8 to 12 MB. The corresponding computational

requirements, $\mathcal{C}_r$, $\mathcal{C}_p$, and $\mathcal{C}_m$ are fixed at 5×10^6 , 1×10^7 , and 5×10^8 CPU cycles, respectively [38].

The communication and computational capabilities of the sensors, UAV, and hive are configured as follows. The CPU frequencies of the sensor f_i , UAV f_u , and hive f_h are set to 0.5, 1.5, and 2 GHz, respectively [39]. The effective switched capacitance coefficients are configured as $\kappa_l = 2 \times 10^{-25}$, $\kappa_u = 2 \times 10^{-26}$ and $\kappa_h = 1.25 \times 10^{-27}$, respectively [40]. For simplicity, we assume that each sensor can achieve a 5 Mbps uplink transmission rate to the UAV with a fixed transmit power of 0.1 W [41], while the UAV is capable of transmitting data to the hive at 30 Mbps using a transmit power of 0.2 W. In addition, all experiments are implemented in Python 3.9 using PyTorch 1.12 and executed on a workstation equipped with an NVIDIA RTX 4060 GPU. The detailed communication and computation parameter settings are presented in Table II.

To facilitate efficient training of the AoT algorithm, the shared encoder is configured with 3 layers, consisting of 8 attention heads and a hidden dimension of 128. For trajectory planning, the model is trained using a batch size of 64 over 600 episodes, where each episode generates 64 candidate trajectories based on a fixed coordinate layout. For task offloading, the output head is implemented as a three-layer MLP with 128, 64, and 3 neurons in each layer respectively [42], with ReLU activations applied between layers. A batch size of 64 is also used for offloading, and the learning rate l_r for both output heads is set as 10^{-4} to ensure stable convergence.

We compare the performance of AoT algorithm with the following benchmark algorithms:

- **Heuristic:** The Artificial Bee Colony (ABC) algorithm is used as baseline for trajectory planning. However, this algorithm is not suitable for task offloading.
- **Value-based DRL:** Dueling Double Deep Q-Network (D3QN) and Double Deep Q-Network (DDQN) are employed solely to solve the task offloading problem.
- **Recurrent Neural Network (RNN):** RNN and its variants, long short-term memory (LSTM) and gated recurrent unit (GRU), are used as alternative encoder modules.
- **BO_ACO:** Bi-Objective Ant Colony Optimization, a state-of-the-art UAV-assisted MEC algorithm that jointly optimizes UAV trajectory and task offloading by balancing energy consumption and computation latency.
- **Traditional Algorithm:** The Greedy algorithm is used to solve the TSP problem, selecting the nearest unvisited cluster as the next hop.

B. Evaluation of AoT Algorithm on Trajectory Planning

1) *UAV Trajectory Visualization:* to intuitively demonstrate the performance of the proposed AoT algorithm in UAV trajectory planning, we visualize the optimized inspection trajectory under varying numbers of sensor clusters. As shown in Fig. 4(a)–(d), the number of clusters increases from 20 to 50. In Fig. 4(a), with 20 clusters, AoT generates a compact path with a total length of 4245.18 meters. As the number of clusters increases to 30 and 40, shown in Fig. 4(b) and Fig. 4(c), the trajectory lengths grow to 4874.85 and 5646.72 meters, respectively. Fig. 4(d) shows the AoT algorithm generates an

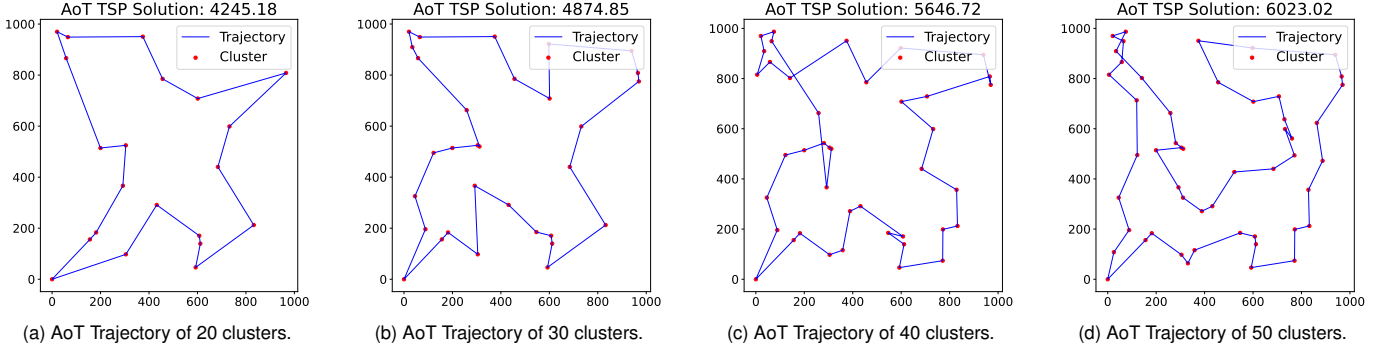


Fig. 4. Trajectory planning results by the AoT algorithm.

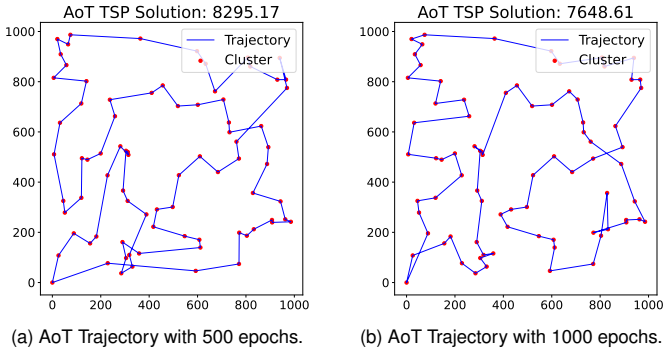


Fig. 5. Trajectory planning under 80 clusters with different training epochs.

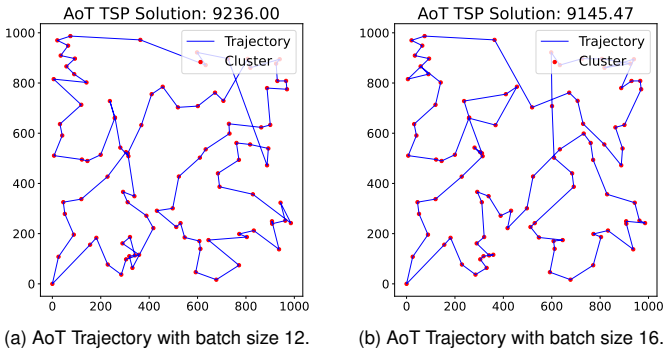


Fig. 6. Trajectory planning under 100 clusters with different batch sizes.

optimal trajectory of 6023.02 meters with 50 clusters. In all, the AoT algorithm performs strong scalability and robustness, proving that the shared-encoder effectively captures global spatial dependencies among clusters.

However, we also observe that as the number of clusters increases, the AoT algorithm tends to produce only near-optimal trajectories. Once the number of cluster exceeds 40, the algorithm occasionally fails to identify the shortest paths between certain clusters. For instance, in Fig. 4(c), small loops appear in the upper-left and central-left regions, and in Fig. 4(d), similar loops emerge in the upper-left area. This is because a low learning rate is required to maintain stable training, while the number of nodes increases, the algorithm may have difficulty escaping local optima since the

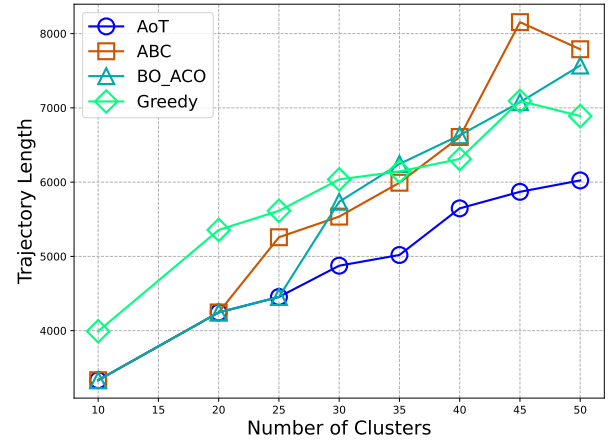


Fig. 7. Trajectory length comparison under different number of clusters.

low learning rate limits its ability to explore more options. Increasing the number of training epochs could help alleviate this issue.

2) *Scalability Analysis*: To evaluate the scalability of the proposed AoT framework, we conduct additional trajectory planning simulations under large-scale scenarios involving 80 and 100 clusters. The results are shown in Fig. 5 and Fig. 6. In Fig. 5, we evaluate the planning performance under 80 clusters with different training epochs. As shown in Fig. 5(a), with a batch size of 18, the AoT algorithm achieves a total path length of 8295.17 meters under 500 epochs, while increasing the epochs to 1000 significantly improves the path quality, reducing the total distance to 7648.61 meters, as shown in Fig. 5(b). Similarly, Fig. 6 shows the results for 100 clusters with 1000 epochs, where training with batch sizes 12 and 16 yields path lengths of 9236.00 meters and 9145.47 meters, as shown in Fig. 6(a) and 6(b), respectively.

Overall, the AoT algorithm demonstrates strong scalability, maintaining effective trajectory planning capabilities across hundreds of clusters. We also note that the primary bottleneck of AoT lies in the memory demand of the Transformer encoder, especially under large batch sizes and cluster numbers. While our current experiments are limited by laptop hardware, we expect that the algorithm will scale even more favorably on devices with larger GPU memory.

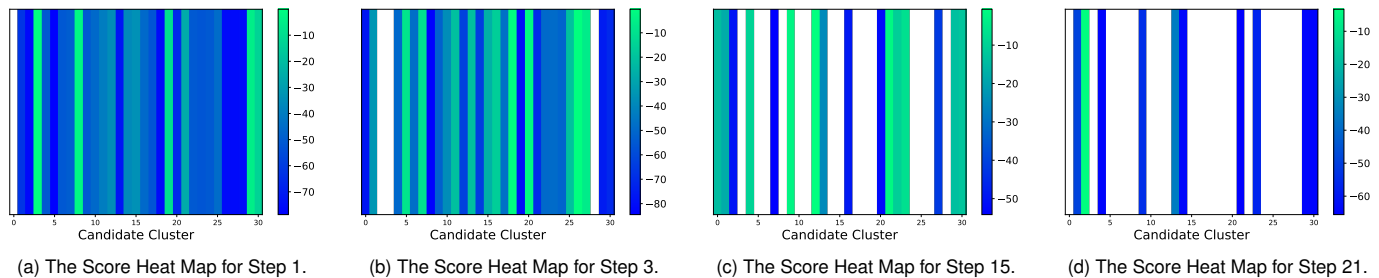


Fig. 8. Heatmap of adjacency scores when the UAV is at different visiting points of the trajectory.

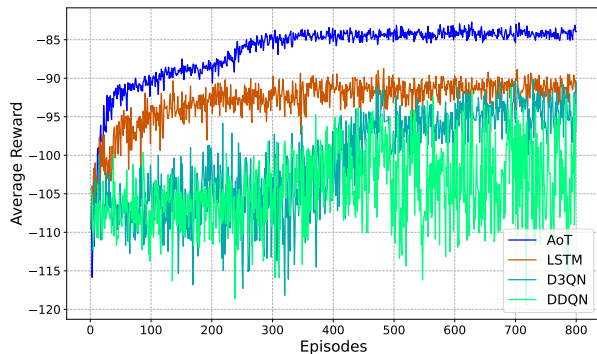


Fig. 9. Convergence performance among different algorithms.

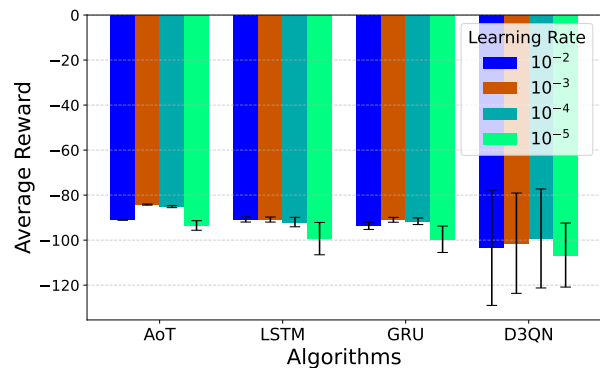


Fig. 10. Reward comparison with respect to different learning rates.

3) *Trajectory Length Comparison*: we also compare trajectory lengths across different algorithms to validate the efficiency of AoT. As shown in Fig. 7, AoT consistently achieves the shortest trajectories as the number of clusters increases. For example, with 45 clusters, AoT generates a trajectory of 5869.52 meters, while the Greedy and BO_ACO algorithms produce longer paths of 7093.39 meters and 7075.64 meters, respectively. The ABC approach results in an even longer trajectory of 8154.82 meters. Compared to these baselines, AoT reduces the total path length by 28.02% over ABC, 14.87% over BO_ACO, and 17.25% over Greedy. This performance gain is mainly attributed to AoT's ability to capture global spatial dependencies via multi-head self-attention and to learn effective planning strategies through DRL. In addition, the Transformer's parallel attention mechanism allows for more flexible and comprehensive exploration of the solution space, especially in complex routing scenarios.

4) *Attention Score over Decision Steps*: to better analyze the selection mechanism of the next visiting cluster, we provide heatmaps of adjacency attention scores at four key steps. Fig. 8 shows the attention distributions in Steps 1, 3, 15, and 21, illustrating the algorithm's focus shifts throughout the trajectory. At step 1 (Fig. 8(a)), the tour begins with a relatively uniform attention distribution, assigning non-negligible probabilities to most candidate clusters. By step 3 (Fig. 8(b)), the attention begins to concentrate on specific clusters. As the tour progresses to step 15 (Fig. 8(c)), a large portion of the clusters has already been visited, and this is reflected by the emergence of many zero-valued entries in the attention matrix. Finally, at step 21 (Fig. 8(d)), the AoT focuses entirely on the few remaining unvisited clusters and makes deterministic

decisions. This stepwise evolution demonstrates that the AoT can dynamically adjust its attention distribution based on the current state. The integration of masking and self-attention enables the AoT to effectively exclude previously visited clusters while maintaining sensitivity to global spatial structure.

C. Evaluation of AoT Algorithm on Task Offloading

1) *Convergence speed*: Fig. 9 shows the average reward curves across 800 training episodes. The AoT algorithm consistently outperforms the baselines, reaching an average reward of -85 around episode 300 and maintaining stable performance thereafter. In contrast, the LSTM-based encoder converges around episode 400. Both the D3QN and DDQN algorithms show even lower final performance, with their average rewards stabilizing below -95. These results highlight the advantage of AoT's self-attention mechanism, which excels at capturing long-range dependencies between tasks and enables efficient parallel processing of task sequences.

2) *Impact of learning rates*: Fig. 10 evaluates the robustness of different models across a range of learning rates, from 10^{-2} to 10^{-5} . The AoT algorithm consistently outperforms all baselines in terms of average reward, with its best performance achieved at $lr = 10^{-3}$, reaching a reward of -84.21. In contrast, LSTM and GRU show moderate sensitivity to learning rate changes, while D3QN performs poorly overall and suffers significant degradation when the learning rate is not properly tuned. The error bars represent the variance of rewards over the final 100 episodes, serving as an indicator of convergence stability. AoT displays the smallest error bars across all settings, demonstrating greater training stability. This advantage

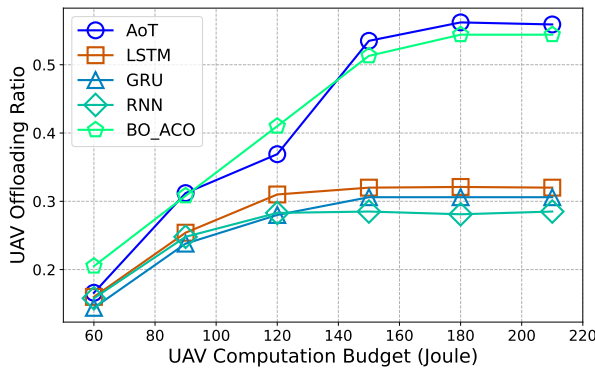


Fig. 11. Task offloading ratio to UAV among different computation budget.

arises from AoT's shared Transformer encoder with multi-head self-attention, which enables parallel token-wise processing and stable gradient propagation, thereby reducing sensitivity to step-size variations during optimization. Unlike sequential models such as LSTM and GRU, the Transformer architecture eliminates recursive dependencies that tend to amplify gradient vanishing or explosion at extreme learning rates.

3) *Impact of UAV computation budget*: Fig. 11 shows a clear positive correlation between the computation budget and the UAV offloading ratio across all algorithms. At every budget level, the AoT algorithm consistently achieves the highest offloading ratio. For instance, with a budget of 180 joules, AoT offloads 56.2% of tasks to the UAV, while BO_ACO, LSTM, GRU, and RNN reach 54.4%, 32.1%, 30.6%, and 28.1%, respectively. This behavior stems from the AoT's ability to jointly capture fine-grained task characteristics and their impact on offloading decisions. The encoder highlights patterns where UAV execution offers a trade-off between delay and energy consumption. Since UAVs provide a middle ground between high communication cost and high computing cost, the AoT model effectively learns to prefer UAV offloading for tasks with moderate data size and computation demands.

VII. CONCLUSION

In this paper, we have proposed an AGI-oriented Transformer (AoT) algorithm to minimize the weighted sum of latency and energy consumption in the UAV-assisted railway inspection networks. An encoder-only architecture has been designed to consider DRL and different output heads for joint trajectory planning and task offloading. The proposed AoT algorithm is a novel lightweight Transformer-based approach. The theoretical innovations in the architectural design and training mechanism are not only tailored for UAV-assisted inspection networks, but also applicable to a wide range of resource-constrained devices for solving multi-objective optimization problems. For the future work, we will design more specialized output heads for handling a wider variety of tasks to further enhance the AoT algorithm.

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